

Capture threshold for electron interacting with circularly polarized waves in parallel static electric fields

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This study investigates the threshold for resonant capture of electrons interacting with circularly polarized electromagnetic waves in a static electric field parallel to the background magnetic field. Captured electrons can channel energy supplied by the static electric field into perpendicular cyclotron motion while maintaining resonance. An analytical expression for the critical wave amplitude required for capture is derived using a nonlinear Schrödinger-type equation, characterizing the transition between passage through resonance and sustained resonant capture. The analytical prediction is compared directly with numerical results, showing strongly agreement. Capture conditions for right- and left-hand circularly polarized waves are also analyzed to examine the influence of wave polarization on resonant electron dynamics.

Key words: Autoresonance, Capture Threshold, Anomalous Doppler resonance, Normal Doppler resonance

1. Introduction

Resonant interactions between electrons and electromagnetic waves govern energy exchange and pitch-angle evolution in magnetized plasmas. Coherent whistler waves can produce substantial, nondiffusive changes in energetic-electron trajectories, with implications for magnetospheric particle dynamics (Yoon & Bellan 2020). In tokamak plasmas, direct observations of runaway-electron-driven whistler waves have established a connection between electromagnetic fluctuations and energetic-electron populations (Spong *et al.* 2018). Externally injected whistler waves have also been proposed as a means of controlling runaway-electron energy through resonant scattering (Guo *et al.* 2018). More recently, experiments in DIII-D demonstrated pitch-angle scattering and suppression of the high-energy electron population following the application of externally launched helicon waves (Choudhury *et al.* 2026). These developments motivate a quantitative understanding of the conditions under which coherent waves can sustain resonant interactions with electrons subject to electric-field acceleration.

In a coherent wave-particle interaction, the relative phase between the wave and electron gyromotion determines whether an electron passes through resonance or remains phase locked. Wave polarization and the Doppler-shifted frequency are therefore central to the accessible resonance and the resulting energy exchange. For propagation parallel to the background magnetic field, the fundamental normal and anomalous Doppler resonances are distinguished by the sign of the Doppler-shifted wave frequency. In the

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anomalous resonance, parallel motion can supply energy to perpendicular gyromotion and the wave. Xu *et al.* (2025) examined this process through angular-momentum conservation and established a correspondence between quantum descriptions and classical simulations of energy transfer. A parallel static electric field adds a further source of energy and changes the electron's parallel momentum, making the onset and persistence of resonant capture sensitive to both the wave polarization and the imposed acceleration.

The underlying single-particle dynamics have a long theoretical history. Roberts & Buchsbaum (1964) obtained exact relativistic solutions for a charged particle interacting with a transverse electromagnetic wave propagating along a uniform magnetic field, including the effects of wave-induced parallel motion and the relativistic variation of the cyclotron frequency. Schram & Beukema (1969) explicitly investigated an additional parallel electrostatic field and identified a transition between sustained average energy growth and passage through resonance, with a transition condition depending on the wave amplitude and refractive index. Related work on gyroresonant surfing showed that an electrostatic force can sustain cyclotron resonance with a monochromatic circularly polarized wave and enable substantial particle energization (Kuramitsu & Krasnoselskikh 2005). These results establish that a parallel electric field can modify the duration and character of resonant energy exchange.

Nonlinear formulations provide further insight into the trajectories associated with strong wave–particle coupling. Bellan (2013) expressed the relativistic dynamics in terms of a pseudo-potential depending on the frequency mismatch, providing a framework for interpreting large pitch-angle changes. Yoon & Bellan (2020) subsequently used this approach to describe nondiffusive scattering of particle distributions by coherent whistler waves. For a right-hand circularly polarized wave in the presence of a constant parallel electric field, Kang *et al.* (2026) demonstrated a reversal of parallel acceleration accompanied by perpendicular energization after resonant trapping. Their study left the exact trapping condition for future investigation. This motivates an explicit analysis of the transition into capture, complementing descriptions of the subsequent trapped motion.

Autoresonance provides a natural framework for examining this transition. In an autoresonant state, nonlinear frequency adjustment allows an oscillator to remain phase locked while the resonance parameters evolve slowly. The principle of sustained synchronism has historical roots in accelerator phase stability (McMillan 1945). Its relevance to electron cyclotron heating was examined by Neishtadt & Timofeev (1987), who studied capture and acceleration in inhomogeneous magnetic traps. Their analysis illustrates that autoresonance can occur at a fixed laboratory wave frequency when particle motion changes the local resonance condition. A broader account of autoresonant excitation and its applications to nonlinear oscillators was presented by Fajans & Friedland (2001).

Autoresonance has also been established in collective plasma dynamics. Experiments on the diocotron mode demonstrated controlled excitation through nonlinear phase locking in non-neutral plasmas (Fajans *et al.* 1999*a,b*). Related approaches have been proposed for exciting large-amplitude Langmuir waves in plasma beat-wave accelerators (Lindberg *et al.* 2004) and for forming and controlling electron phase-space holes using a chirped driving wave (Friedland *et al.* 2006). These studies involve different resonances and geometries, but share the competition between an evolving frequency mismatch and a nonlinear response capable of maintaining phase locking.

A central result of autoresonance theory is the existence of a finite drive threshold for capture during resonance passage. For a broad class of weakly nonlinear oscillators with a linear frequency sweep and sufficiently small initial amplitude, the critical drive scales as the three-quarter power of the sweep rate (Fajans & Friedland 2001). Friedland (2008)

formulated the capture dynamics using a driven nonlinear Schrödinger-type equation and examined how the threshold depends on the initial point of the resonance passage. Further studies quantified the influence of damping (Fajans *et al.* 2001) and showed how thermal fluctuations broaden the capture transition while self-fields modify its width (Barth *et al.* 2009). The conversion of a normalized threshold into a physical wave-amplitude criterion consequently requires explicit specification of the reduction, normalization and initial conditions.

For electrons interacting with a monochromatic wave in uniform parallel static fields, the electric field changes the parallel momentum and hence the Doppler-shifted wave frequency. This produces an effective detuning sweep despite the fixed laboratory wave frequency. Connecting this configuration to autoresonance requires a reduction that relates the physical field parameters and wave polarization to the coefficients of the capture equation. Such a connection provides a means of determining when the wave-induced nonlinear response can compensate the evolving detuning and sustain resonant capture.

The present study develops this connection for electrons interacting with a circularly polarized electromagnetic wave propagating along a uniform background magnetic field in the presence of a parallel static electric field. The resonant dynamics are reduced to a nonlinear Schrödinger-type equation, yielding an analytical criterion for the critical wave amplitude under the specified approximations and initial conditions. Direct comparison with numerical results shows close agreement. Capture conditions for right- and left-hand circularly polarized waves are examined, with particular attention to anomalous Doppler capture and the transfer of energy supplied by the static electric field into perpendicular cyclotron motion. The resulting analysis connects established autoresonance theory to a physical electron-capture criterion and clarifies the transition from resonance passage to sustained phase locking.

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1.2. *Lists*

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- (5) List entry five.

- (i) List entry one.
- (ii) List entry two.
- (iii) List entry three.
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- (v) List entry five.

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$$AB = \sum_i a_i b_i$$

2. FIRST-ORDER HEADING

Fourth-Order Heading

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Appendix
First order heading

TABLE 3. Table caption table caption table caption table caption table caption table caption
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Column One	Column Two	Span Col 3 & 4		Span Col 5 & 6	
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6-31+G(d,p)	460	2.12	1.04	8.63	3.48
6-31++G(d,p)	476	2.24	0.94	9.22	3.73
6-31++G(df,p)	676	3.71	1.67	16.36	6.00
6-311+G(d,p)	572	2.86	1.29	11.90	4.56
6-311++G(3df,2p)	1,060	7.54	3.25	36.70	11.45

^{a)}Table Footnote
^{b)}Second Table Footnote.